

For questions 1-2, consider the monomial $-\frac{1}{8}x^2y^2$ and the linear expression $(12x - 7)$.

1. Determine the product of $-\frac{1}{8}x^2y^2$ and $(12x - 7)$ using an area model.

$12x$	-7
$-1.5x^3y^2$	$\frac{7}{8}x^2y^2$

$-\frac{1}{8}x^2y^2$

$= -1.5x^3y^2 + \frac{7}{8}x^2y^2$

2. Determine the product of $-\frac{1}{8}x^2y^2$ and $(12x - 7)$ by applying the distributive property.

$$-\frac{1}{8}x^2y^2(12x - 7) = -1.5x^3y^2 + \frac{7}{8}x^2y^2$$

For questions 3-4, consider the monomial $\frac{3}{5}s^3k$ and the linear expression $(\frac{1}{3}s - 15)$.

3. Find the product of $\frac{3}{5}s^3k$ and $(\frac{1}{3}s - 15)$ using your preferred method.

$$\frac{3}{5}s^3k\left(\frac{1}{3}s - 15\right) = \frac{1}{5}s^4k - 9s^3k$$

4. Write another expression equivalent to the product.

Answers may vary.

$$-9s^3k + \frac{1}{5}s^4k$$

For questions 5-10, find each product using your preferred method.

5. $1.5rg^2(0.25 + 0.8r)$

$$0.375g^2r + 1.2g^2r^2$$

6. $-5.25m^5(4m - 1.5)$

$$-21m^6 + 7.875m^5$$

7. $\frac{2}{3}q^3n^2\left(\frac{1}{5}q + \frac{2}{7}\right)$

$$\frac{2}{15}n^2q^4 + \frac{4}{21}n^2q^3$$

8. $\left(6w - \frac{1}{4}\right)\left(-\frac{9}{2}z^5\right)$

$$-27wz^5 + \frac{9}{8}z^5$$

9. $(4.02 - b)(b^3cx^4)$

$$4.02b^3cx^4 - b^4cx^4$$

10. $-12f^4n^4\left(\frac{4}{3}n - 8\right)$

$$-16f^4n^5 + 96f^4n^4$$